

JavaScript SEO and Rendering

Google will eventually see your JavaScript. Several AI engines never will.

Bottom line

Google indexes in two waves: raw HTML first, then JavaScript execution 24 to 72 hours later. That delay is survivable. What is not survivable is that AI crawlers frequently skip JavaScript entirely, so a client-rendered site can be perfectly visible in Google and structurally invisible to ChatGPT and Perplexity.

The two-wave model

Wave one. Googlebot fetches the raw HTML and indexes what is in it, immediately.

Wave two. The URL is queued for the Web Rendering Service, which runs JavaScript and indexes what appears. This happens **24 to 72 hours later**, sometimes longer under load. [practitioner]

Consequences:

- Content only present after JavaScript execution is invisible for days
- Links only present after execution are not followed until wave two, delaying discovery of everything behind them
- Time-sensitive content can be stale before it is indexed
- Any wave-two failure means that content is never indexed at all

The part that changed

AI crawlers often do not execute JavaScript at all.

GPTBot, OAI-SearchBot, ClaudeBot and PerplexityBot crawl independently of Googlebot, and several of them fetch raw HTML and stop. There is no wave two. [practitioner]

This is the single most consequential technical fact in the course for 2026. A React or Vue site with client-side rendering can rank normally in Google, because Google eventually renders, and be **completely absent from AI answers**, because the AI crawler saw an empty shell.

You would never detect this by checking your Google rankings.

The rendering options

Client-side rendering (CSR). The browser receives a near-empty HTML shell plus JavaScript that builds the page. Worst case for SEO and effectively fatal for AI visibility.

Server-side rendering (SSR). The server renders complete HTML per request. Bots receive real content immediately. Best for both Google and AI crawlers.

Static site generation (SSG). Pages pre-rendered at build time and served as static HTML. Fastest and excellent for SEO. Ideal for content that does not change per request.

Incremental static regeneration (ISR). SSG that revalidates on a schedule. A good middle ground for content sites.

Dynamic rendering. Serve pre-rendered HTML to bots and the JS app to users. Google previously described this as a workaround rather than a recommendation, and it adds complexity and a maintenance burden. Prefer SSR or SSG.

The default recommendation: if content needs to be found, render it server-side or statically. Modern frameworks make this the default path rather than an exotic one, which was not true a few years ago.

What breaks in practice

Content behind interaction. Tabs, accordions and "load more" that fetch content on click. If it requires a click, bots do not see it. Content in a tab already present in the DOM and hidden with CSS is fine.

Infinite scroll with no paginated fallback. Items past the initial load are unreachable.

JavaScript-only links. `<span onclick="navigate()">` is not a link. Bots follow `<a href>`. This is the most common cause of whole sections going undiscovered.

Content loaded from an API after page load with no server-rendered fallback.

Blocked JavaScript files. From Section 21, blocking your JS in robots.txt means the render fails.

Slow or failing JS. If a script times out during rendering, that content is not indexed. Bots do not retry patiently.

How to test

Google's URL Inspection, Test live URL, View tested page. This is the ground truth for what Google renders. Check both the screenshot and the HTML tab.

Compare raw against rendered. View source shows raw HTML that a non-rendering crawler sees. Inspect element shows the rendered DOM. **The gap between them is what AI crawlers miss.**

Disable JavaScript in your browser and load the page. Chrome DevTools, Command Palette, "Disable JavaScript". What remains is roughly what a non-rendering bot receives. This is the fastest

and most sobering test available.

Fetch the raw HTML directly with `curl` or any HTTP client and search it for your main heading and body text. If they are absent, so is your AI visibility.

The pragmatic fix path

You rarely need to rebuild.

- 1 **Server-render or statically generate the content that must be found.** Usually main copy, headings, links and metadata. Interactive components can stay client-side.
- 2 **Make navigation real** `<a href>` **links.** Cheap and high-impact.
- 3 **Provide paginated fallbacks** behind infinite scroll.
- 4 **Move content out of click-gated interactions**, or ensure it is in the DOM and hidden with CSS rather than fetched on demand.
- 5 **Ensure metadata is server-rendered.** Title, meta description and canonical injected by JavaScript are unreliable and frequently missed.

That last one catches people constantly. A client-side SEO plugin setting titles after load often means bots see the default template title.

Why this matters

this is the failure mode most likely to be silently costing you AI visibility right now while your Google rankings look fine. The diagnostic takes two minutes, disable JavaScript and reload, and almost nobody runs it.

Do this now

- 1 **Open your most important page and disable JavaScript.** Reload. What is still there?
- 2 **View source and search for your H1 text.** Present in the raw HTML, or not?
- 3 **Do the same for your main body copy, your title tag, and your canonical tag.**
- 4 **Run URL Inspection, Test live URL, View tested page** on the same URL. Compare Google's rendered version against raw source.
- 5 **Check your navigation links** in raw HTML. Are they real `<a href>` elements?
- 6 **Fetch the page with curl** and `grep` for a distinctive sentence from your content. This is exactly what a non-rendering AI crawler sees.
- 7 **Write down what is missing from the raw HTML.** That list is your AI visibility gap.
- 8 **If content is missing, identify what your site would need** to server-render it, and note it as a development task with a clear justification.

Capstone step

You know precisely what your site looks like without JavaScript, which is what several AI crawlers see. Any gap between raw and rendered HTML is documented as a specific development task rather than a vague concern.

Key takeaways

- Google renders JavaScript in a second wave 24 to 72 hours later. Survivable, but it delays everything.
- AI crawlers frequently do not execute JavaScript at all. A client-rendered site can rank fine in Google and be entirely absent from AI answers.
- Disable JavaScript and reload. What remains is roughly what a non-rendering bot sees, and it is the fastest diagnostic in this tier.
- Server-render the things that must be found: content, headings, real `<a href>` links, and especially metadata.